voice coil with the frame of the speaker and so as vibration will be proper.

When the amplified audio signal is given to the voice coil, it will create variable magnetic field. The force develops in between the static and dynamic magnetic field move the voice coil front and back. Due to this the paper cone gets vibrated.

2.4.2.2 Impedence of the Speaker

Speakers are classified (as 4 ohms, 8 ohms ,16 ohms) based on their impedence rating. The impedence rating is depends upon the thickness and numbers of turns of the voice coil. For proper matching the audio frequency amplifier output must be equal to the voice coil Impedence. If it is not equal, then it is said to be improper matching. This improper matching affects the sensitivity of the audio output.

2.4.2.3 Frequency Handling of Speakers

Speakers are classified as per the frequencies handled by them. They are

1. Woofer
2. Squawker
3. Tweeter



Figure 2.27
Types of Speakers

A Woofer

B Squawker

C Tweeter

Woofer (Low-Frequency range): The speaker which functions efficiently in the low frequency (20 Hz to 1000 Hz) audio signal range is called as Woofer and is shown in Figure 2.27(a). Since a large amount of power is needed to deliver this range, these types of speakers are designed as big in size and diameter. Presently, sub-woofer with frequency range of 20-200 Hz is also widely used.

Squawker (Mid-Frequency range): The speaker which functions efficiently in the mid frequency (2 to 8 kHz) audio signal range is called as Squawker or mid-range speaker and is shown in Figure 2.27(b).

Tweeter (High-Frequency range): The speaker which functions efficiently in the high frequency (8 to 20 kHz) audio signal range is called as Tweeter. It has less weight and small in size. Now, Piezo tweeters are also used. Tweeter is shown in Figure 2.27(c).

2.4.2.4 Two Way and Three-way Speaker System

A single speaker cannot deliver the full range of the audio signal (20 Hz to 20 kHz) with good quality and required fidelity. So, by using cross-over network with Woofer, Squawker and Tweeter, the full audio range can be delivered with good quality and fidelity. This is done in two channels (ways), for that, two speaker boxes are used. This is called as stereophonic system. If all the three speakers are used in one speaker box, it is said to be monophonic system (one channel).

2.4.3 Wattage Rating of Speakers

Wattage rating of speakers is nothing but the capability of voice coil which delivers the power of sound signal. Speaker having

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